

Demonstrating check-derivative with w_3 and scalar values.

original-value- $w_3 = 4$

increase- $w_3 = 6$

do forward prop

actual = 10

$$\text{predicted} = 6 \cdot \frac{3}{2} = 9$$

$$\text{loss} = \frac{1}{2} (10 - 9)^2 = \frac{1}{2}$$

decrease- $w_3 = 2$

do forward prop

actual = 10

$$\text{predicted} = 2 \cdot \frac{3}{2} = 3$$

$$\text{loss} = \frac{1}{2} (10 - 3)^2 = \frac{1}{2} (7)^2 = \frac{49}{2}$$

$$\text{numerical-loss} = \frac{\frac{49}{2} - \frac{1}{2}}{4} = \frac{\frac{48}{2}}{4} = \frac{24}{4} = 6$$

revert to original

do forward and back prop

$$\text{predicted} = 4 \cdot \frac{3}{2} = 6$$

$$dL - dw_3 = 10 - 6 = 4$$

$$dL - dw_3 = 4 \cdot \frac{3}{2} = 6$$